the side chains of amino acids in a polypeptide are known together as the back bone or main chain of the polypeptide.

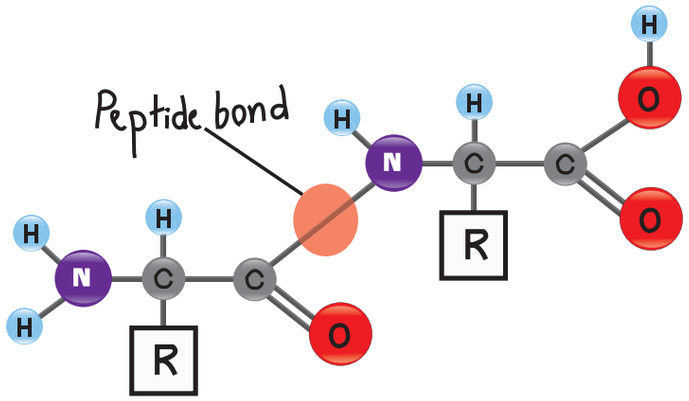


Figure 3.8 Chemical structure of a dipeptide and a peptide bond

Peptide bonds have some important properties, which are

i. Peptide bonds are generally in *trans* conformation. However in rare conditions peptide bonds formed by proline can adopt a *cis* conformation.

ii. Peptide bonds have a partial double bond character, which gives them a planar nature and hence cannot be rotated.

iii. Since peptide bonds are amide linkages the –C=O and –NH groups cannot donate or accept protons and are uncharged. The net charge of a polypeptide can come only from the N terminus amino group, C terminus carboxyl group and the side chains of the amino acids.

iv. Despite not being ionisable, the –C=O and –NH groups of peptide bonds are polar and can involve in the formation of hydrogen bonds. This property is important for the formation of secondary structures of proteins.

3.3.2 Secondary structure of proteins

The back bone of a polypeptide forms regular structural arrangements by making hydrogen bonds with its neighbouring amino acids. As a rule, these hydrogen bonds are always between the main chain –NH group and –C=O group. There are three main types of secondary structures present in proteins namely α Helix, β sheet and β turn.

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